Uncertainty-Aware Portfolio Optimisation Using Machine Learning

Week 3 Problem Set Report

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1 Classification

1.1 1.1 Logistic Regression for Credit Default Prediction

Summary: To anticipate financial distress within a two-year horizon, a binary logistic regression model was implemented using the Give Me Some Credit dataset. Missing entries were addressed via median imputation, and continuous features underwent Z-score standardization utilizing scaling parameters derived solely from the training partition.

Table 1: Logistic Regression Test Performance

Metric	**Value**
Accuracy	0.9338
Precision	0.5593
Recall	0.0439
F_1-Score	0.0814
AUC-ROC	0.7151

Coefficient Interpretation: Evaluating the top five features ranked by absolute coefficient magnitude highlights *NumberOfTime60-89DaysPastDueNotWorse* (3.768), *NumberOfTime30-59DaysPastDueNotWorse* (2.038), *NumberOfTimes90DaysLate* (1.890), *age* (0.404), and *MonthlyIncome* (0.403). The stark dominance of past delinquency metrics over baseline demographic indicators suggests that recent histories of severe payment delays serve as exponentially stronger signals for future default risk than traditional variables like age or income.

1.2 1.2 Decision Tree Classifier with Pruning

Summary: Using the Heart Disease dataset, decision trees were evaluated to inspect how maximum depth restrictions and cost-complexity pruning mitigate overfitting. The unconstrained baseline framework overfitted the data severely, delivering a perfect 1.0000 training accuracy but dropping to 0.6522 on the test set.

Pruning Analysis: Enforcing cost-complexity regularization ($\alpha^* = 0.0312$) improved generalization, yielding a test AUC-ROC of 0.7000. By penalizing structural

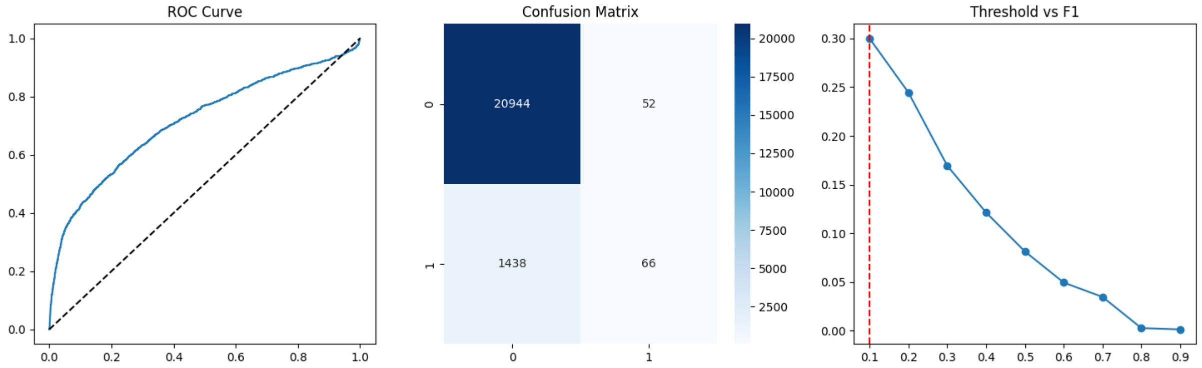

Figure 1: Logistic Regression Diagnostic Plots: ROC, Confusion Matrix, and Threshold Optimization

Table 2: Decision Tree Strategy Comparison

Strategy	**Train Accuracy**	**Test Accuracy/AUC**	**Parameter**
Unpruned	1.0000	0.6522 (Acc)	Depth = None
Cost-Complexity Pruned	-	0.7000 (AUC)	$\alpha^* = 0.0312$

complexity as a function of terminal leaf count, this pruning mechanism successfully eliminated noisy, high-variance branches fitted to training-set anomalies, resulting in an easily interpretable tree structure with superior predictive power.

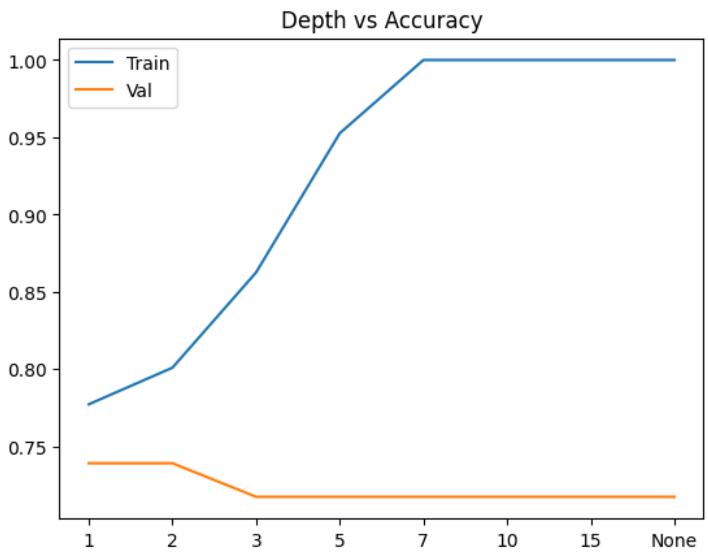

Figure 2: Training vs Validation Accuracy across Maximum Tree Depths

2 Regression

2.1 2.1 Regularised Regression for House Prices

Summary: To navigate severe multicollinearity within the Ames housing dataset, Ridge, Lasso, and Elastic Net regression techniques were benchmarked using 5-fold cross-validation.

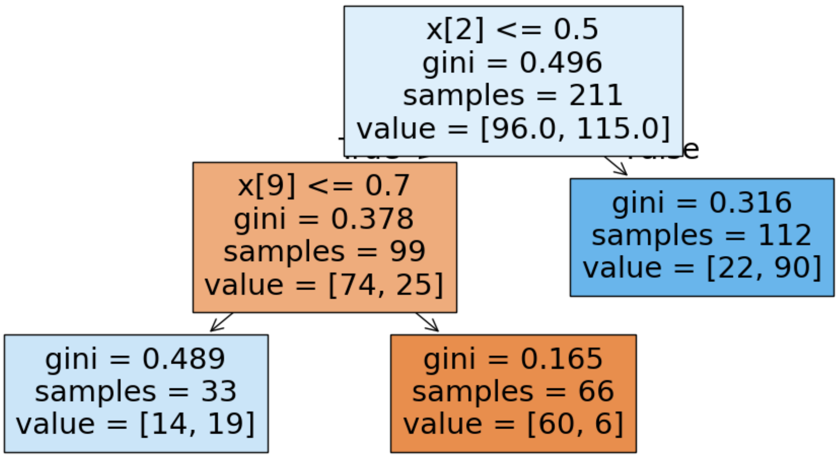

Figure 3: Cost-Complexity Pruned Structural Tree Architecture

To stabilize conditional variance and satisfy homoscedasticity assumptions, the target variable was log-transformed via $y \leftarrow \log(1 + y)$.

Table 3: Regularised Regression Test Set Comparison

Model	**Optimal** λ^*	**Active Features**	**Test RMSE**	R^2
Ridge	429.1934	227	27,393.03	0.8878
Lasso	0.0054	81	26,721.40	0.8917
Elastic Net	0.0687	74	28,203.20	0.8877

Geometric Interpretation of Sparsity: The sparsity intrinsic to Lasso stems directly from its L_1 norm regularization, which shapes the parameter constraint region into a sharp hyperdiamond. As the elliptical contours of the quadratic loss function expand from the OLS solution, they tend to first intersect this constraint budget at its vertices, where multiple coefficient coordinates are driven precisely to zero. This automated feature selection mechanism proves highly advantageous in noisy financial environments by filtering out collinear background noise and highlighting true foundational drivers.

2.2 2.2 Locally Weighted Regression (LWR)

Summary: A non-parametric Locally Weighted Regression model was built from scratch using a Gaussian kernel function to map the non-linear dynamics between neighborhood median income levels and home values.

LWR vs OLS Asymptotics: The optimal smoothing parameter was identified at $\tau = 1.0$, surpassing the performance of the rigid OLS baseline. Analytically, as the bandwidth parameter approaches infinity ($\tau \to \infty$), the Gaussian weights $w_i = \exp(-\frac{||x_i - x_q||^2}{2\tau^2})$ converge uniformly to 1 for all data coordinates. Consequently, the localized diagonal weight matrix W scales directly into the identity matrix I, causing the localized estimator to mathematically collapse back into the global ordinary least squares solution $\hat{\theta} = (X^\top X)^{-1} X^\top y$.

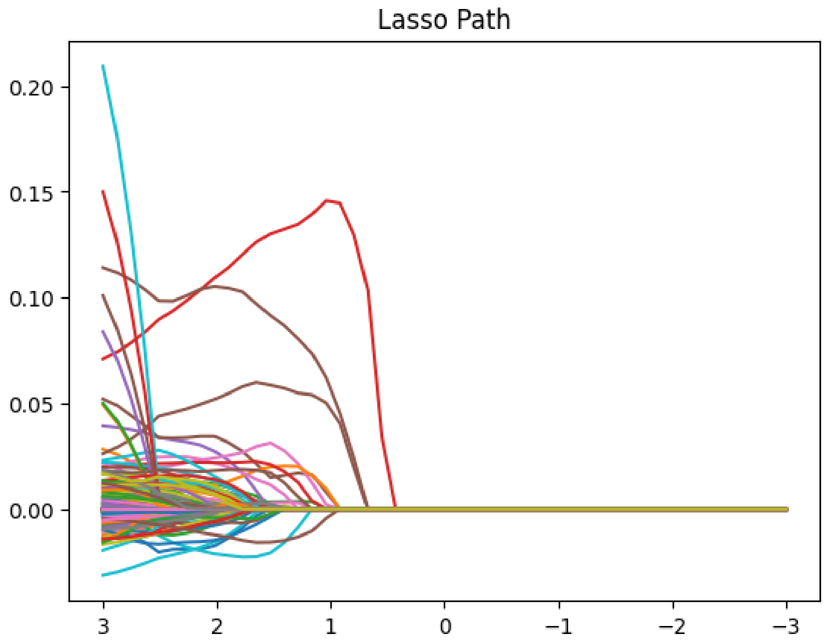

Figure 4: Lasso Regularization Parameter Path Evolution

Table 4: LWR Bandwidth (τ) vs. Ordinary Least Squares (OLS)

Bandwidth (τ)	**Test RMSE**
0.05	95,370.02
0.10	102,821.70
0.30	92,454.65
1.00	91,891.47 **(Best)**
3.00	92,165.57
OLS Baseline	92,238.61

3 Ensemble Methods

3.1 3.1 Random Forest for Fraud Detection

Summary: A Random Forest model was trained on an undersampled subset ($n = 50,000$) of a financial fraud dataset. Inverse class-frequency weighting was applied to counter severe label asymmetry.

- **Out-of-Bag (OOB) Accuracy:** 0.9728
- **Test AUC-ROC:** 0.8833

Feature Importance Discrepancies: Mean Decrease in Impurity (MDI) systematically inflates the importance of continuous or high-cardinality predictors due to its mathematical bias toward features that permit a greater number of unique split points. Conversely, Permutation Importance mitigates this bias by measuring accuracy degradation on an independent validation fold when a feature's values are randomly shuffled. This breaks structural associations with the target and provides a cleaner estimate of true generalization value.

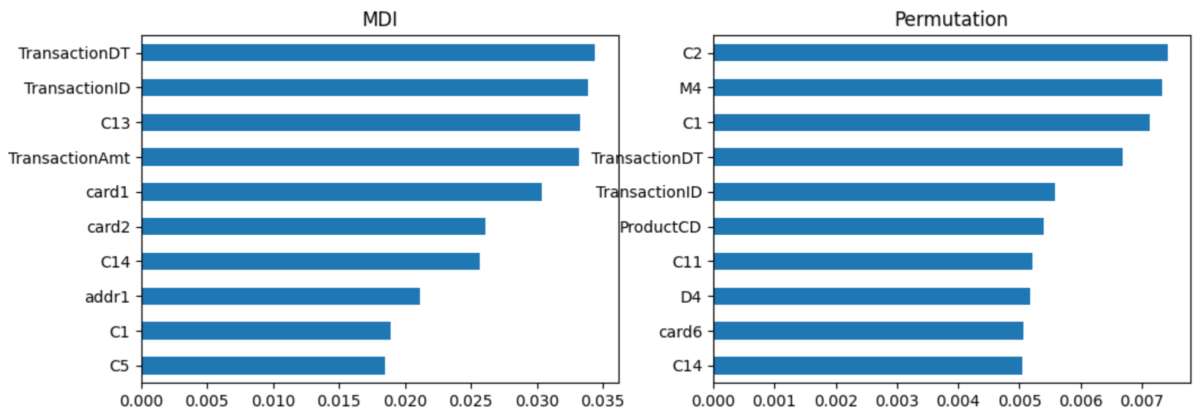

Figure 5: Comparison of Mean Decrease in Impurity (MDI) vs Permutation Importance

Class Imbalance Metrics: Under highly skewed class distributions, evaluating performance via the Precision-Recall curve (AUC-PR) is considerably more robust than relying on the AUC-ROC metric. Because AUC-ROC incorporates True Negatives within its denominator, its scores can be heavily inflated by an overwhelming majority class. AUC-PR targets the minority (fraud) class explicitly, offering a more dependable diagnostic of precision-recall tradeoffs.

3.2 3.2 XGBoost for Stock Return Classification

Summary: An XGBoost framework was configured to predict binary 21-day forward equity returns using standard technical indicators (such as RSI and rolling volatility). A chronological train-test split was instituted (Train ¡ 2017, Test $\geq$ 2017) to prevent temporal leakage.

- **Test AUC-ROC:** 0.5002

- **Test AUC-PR (AP):** 0.5671

Look-Ahead Bias in Cross-Validation: Standard randomized k-fold cross-validation breaks down when applied to financial time series since shuffling data points destroys chronological dependencies. This architectural mistake introduces future information back into past training cycles (e.g., conditioning a model on 2018 regimes to predict 2017 events), leading to look-ahead bias. The resulting performance metrics are artificially inflated and mask poor generalization capabilities in practical deployment.

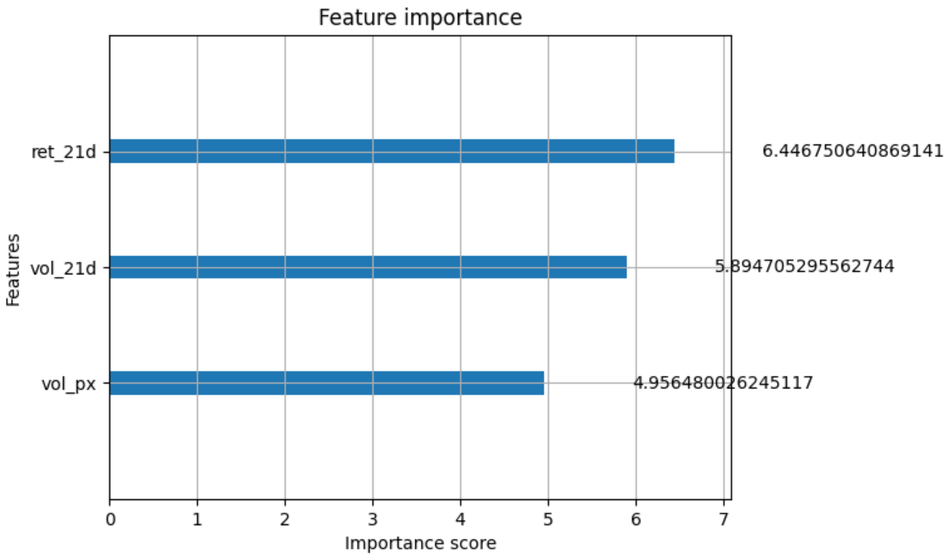

Figure 6: XGBoost Feature Importance Vector (Information Gain Scale)

4 Bias-Variance Analysis

4.1 4.1 Empirical Bias-Variance Decomposition

Summary: Utilizing a bootstrapping setup ($B = 50$), the out-of-sample expected prediction error was empirically factored into its squared Bias ($Bias^2$) and Variance constituents across several distinct algorithms.

Table 5: Bias-Variance Tradeoff Comparison

Model	Bias^2	Variance	Total Error
Logistic Regression	0.1445	0.0035	0.1480
Decision Stump	0.1652	0.0175	0.1828
Unpruned Tree	0.1504	0.1457	0.2961
Random Forest	0.1499	0.0070	0.1569

Bagging vs. Boosting Explanation: Bagging (e.g., Random Forest) operates by dampening the variance of complex, unpruned base learners through uniform prediction averaging. As defined by the analytical variance formula $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$, expanding the ensemble size B drives the second term toward zero, though it leaves the inherent structural bias of the single base estimators unadjusted. Conversely, Boosting iteratively constructs an ensemble of shallow trees focused on minimizing the residual errors of prior iterations. This sequential correction directly compresses model bias, while the small depth restrictions and small learning rates preserve tight control over ensemble variance.

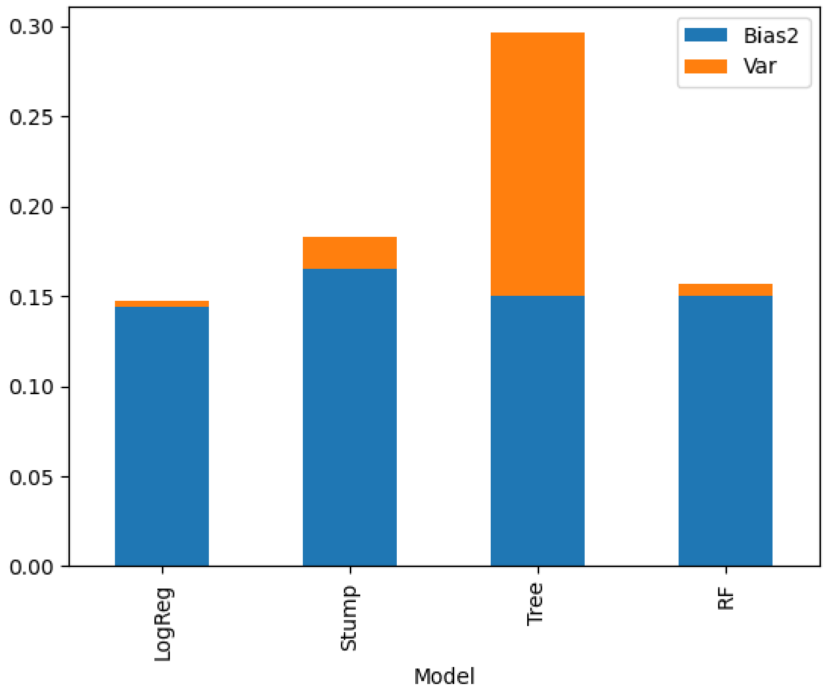

Figure 7: Empirical Error Analysis: Stacked Bias Squared vs Variance Profile

5 Gradient Boosting in a Portfolio Context

5.1 5.1 Gradient Boosting with Uncertainty Estimation for Return Prediction

Summary: A specialized Gradient Boosting ensemble structure was engineered to forecast 21-day forward returns using 454,074 rows extracted from historical NIFTY-50 data. An expanding time-series window validation layout was applied to strictly preserve chronological causality.

Table 6: Expanding-Window CV Optimal Parameters

Hyperparameter	**Optimal Value**
Learning Rate	0.05
Max Depth	3
Number of Estimators	100
Test AUC-ROC	**0.5214**

Uncertainty Estimation via Ensemble Disagreement: Model uncertainty was quantified by training an ensemble composed of $M = 20$ independent gradient boosting architectures across unique bootstrap samples of the training data. For each target out-of-sample instance, the expected probability ($\overline{p}_i$) and corresponding cross-ensemble standard deviation (σ_i) were tracked. The accompanying scatter plot maps areas across the probability range where the sub-models express peak disagreement.

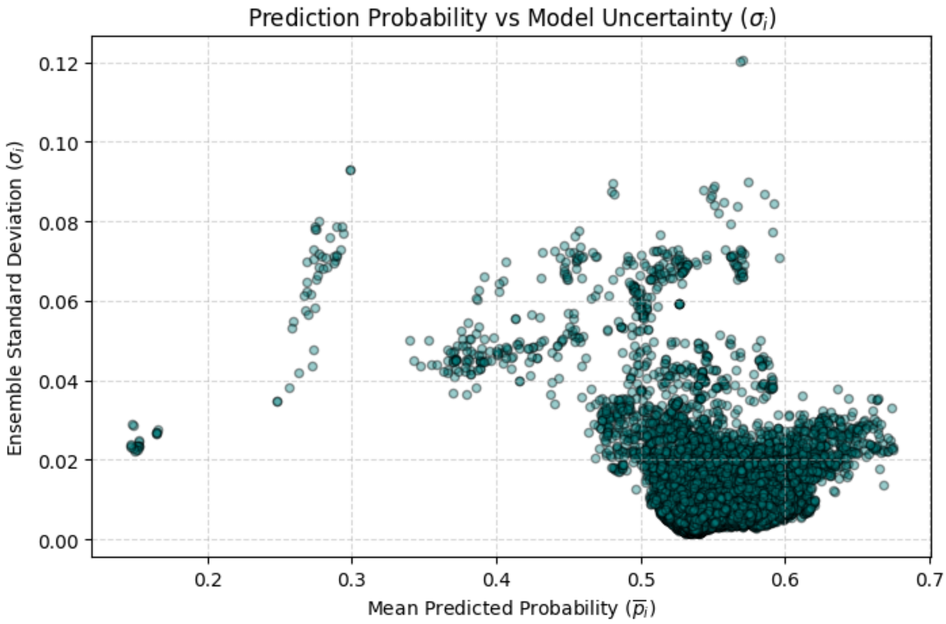

Figure 8: Prediction Probability ($\overline{p}_i$) vs Model Uncertainty (σ_i)

Reflection on Live Trading Risks: Deploying this gradient boosting configuration in live market operations involves substantial structural hazards. First, financial time series exhibit high non-stationarity; structural market regime shifts can occur rapidly, causing algorithms tuned on historical dependencies to overfit to relationships that vanish during macroeconomic disruptions. Second, look-ahead bias remains a major vulnerability, where minor calculation oversights—such as leaking future info into rolling indicators—can produce deceptively flawless historical simulations. Third, operational transaction costs, slippage, and bid-ask spreads stemming from a high-turnover long-short decile trading strategy can quickly erode the modeled alpha. Finally, managing these factors demands an uncertainty-aware approach to portfolio position sizing. Test points yielding elevated σ_i scores point to significant disagreement among the ensemble sub-models; instead of assigning uniform capital weights, an active portfolio framework should scale allocations inversely to σ_i, reducing risk exposure on highly volatile model predictions to safeguard capital.